

2 marks

Cloud Computing

1) Define Platform as a Service (PaaS) with one example.

Solution:

Platform as a Service (PaaS) is a cloud computing model that provides a complete environment for developing, running, and managing applications ~~with~~ without the need to build or maintain the ~~under~~ underlying infrastructure.

An example of PaaS is Google App Engine, which allows developers to build and deploy scalable web and mobile apps on a fully managed platform.

2) What is Open Virtualization Format.

Solution:

Open Virtualization Format (OVF) is an open standard for packaging and distributing virtual machines (VMs).

It provides a platform-independent and portable way to package virtual appliances, using XML to describe their configuration and metadata, which facilitates easy deployment and mobility across different virtualization platforms.

3) List any two features of SQL and Azure.

Solution:

• Features of SQL -

(i) Data Integrity and ACID Compliance: SQL databases enforce data integrity through constraints and follow the ACID (Atomicity, Consistency, Isolation, Durability) to ensure reliable transactions and data consistency.

(ii) Structured Data and Query Language: SQL uses a predefined schema to store data in a structured, tabular format, organizing data into tables, rows, and columns.

• Features of Microsoft Azure

- (i) Scalability and Elasticity: Azure provides both vertical and horizontal scalability, allowing users to dynamically adjust resources based on demand.
- (ii) Built-in Security and Compliance: Azure offers robust, layered security features, including multi-factor authentication, data encryption, advanced threat detection, and comprehensive auditing tools.

4) What is mean by Cloud Bursting.

Solution:

Cloud bursting is a hybrid cloud deployment model where an application runs in a private cloud or on-premises data center, and "bursts" into a public cloud to use additional resources when demand spikes. This allows organizations to handle workload surges without overspending on their own infrastructure, paying only for the public cloud resources when they are needed.

5) Name any two types of ddos attack.

Solution:

Two types of ddos attack is —

- (i) Volumetric Attacks: It aims to consume all available network bandwidth with a massive flood of traffic, making the service inaccessible to legitimate users.
- (ii) Application Layer Attacks: It targets specific features or functions of a web application to consume server resources, such as CPU and memory, making the application slow or crash.

6) Define load balancing.

Solution:

Load balancing is the process of distributing network traffic and workloads across multiple computing resources, such as servers, to ensure no single resource is overwhelmed. This method improves

application performance, reliability, and scalability by distributing requests evenly, which minimizes latency, increases throughput, and enhances availability.

7) Define Service Oriented Architecture (SOA).

Solution:

SOA is a design paradigm where applications are built using self-contained, reusable software components called "services" that can be accessed and combined over a network. This approach allows for more flexible, scalable, and efficient development of cloud applications by enabling services to be developed and managed independently, yet interact seamlessly across different technologies.

8) Define cloud computing as per the NIST model.

Solution:

According to the National Institute of Standards and Technology (NIST), Cloud Computing is defined as:

"A model for enabling ubiquitous, convenient, ~~on-demand~~ on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction."

9) List any two advantages of cloud computing over traditional computing.

Solution:

Two advantages of cloud computing over traditional computing:

(i) Scalability and Flexibility — Cloud resources can be easily scaled up or down based on user demand, unlike traditional systems that require

manual hardware upgrades.

(ii) Cost Efficiency — Cloud computing follows a pay-as-you-go model, reducing the need for large upfront investments in hardware and maintenance costs associated with traditional infrastructure.

10) What is hypervisor. Name two example.

Solution:

A hypervisor, also known as a virtual machine monitor (VMM), is a software, firmware or hardware layer that enables a single physical computer to run multiple independent virtual machines simultaneously.

Two examples are:

(i) Microsoft Hyper-V

(ii) Oracle VM Virtual Box

11) Mention two features of Google App Engine.

Solution:

(i) Automatic Scaling: Google App Engine automatically scales applications up or down based on the amount of traffic or requests, ensuring optimal performance without manual intervention.

(ii) Managed Infrastructure: It provides a fully managed environment where Google handles server management, load balancing, patching, and monitoring, allowing developers to focus only on writing and deploying code.

12) State the ^{CAP} theorem in the context of cloud database system.

Solution:

The CAP theorem states that a distributed database system in a cloud environment can only guarantee two of the following three properties at any given time: Consistency, Availability and Partition Tolerance. When a network partition occurs, the system must choose between maintaining Consistency or Availability.

13) Name any two common cloud deployment model.

Solution:

Two common deployment models are:

(i) Public Cloud: Here, the computing services and infrastructure are provided by third-party ~~for~~ providers like AWS, Microsoft Azure over the public internet and are shared among multiple organizations.

(ii) Private Cloud: It is dedicated exclusively to a single organization. It can be hosted on-premises within a company's data center or managed by a third-party provider off-site.

5 marks

1) Differentiate between Grid Computing, Cluster Computing and Cloud Computing.

Solution:

Feature	Grid Computing	Cluster Computing	Cloud Computing
(i) Definition	A distributed computing model where geographically dispersed, heterogeneous systems work together to solve large computational problems.	A group of closely connected and homogeneous computers working together as a single system, usually within the same location.	A service-oriented computing model that provides on-demand access to computing resources over the internet.
(ii) Resource Ownership	Resources belong to multiple organizations but shared collaboratively.	Resources are owned and managed by a single organization.	Resources are owned and controlled by cloud providers.
(iii) Scalability	Highly scalable.	Limited scalability.	Extremely scalable.
(iv) Cost	Typically lower cost by using existing distributed system.	Cost depends on maintaining hardware within the organization.	Pay-as-you-go model, reducing upfront infrastructure cost.

Feature	Grid Computing	Cluster Computing	Cloud Computing
(v) Application Type	Scientific research, complex mathematical modeling, weather forecasting.	High performance tasks like parallel processing, database servers, scientific labs.	Web hosting, enterprise applications, storage, AI/ML workloads.
(vi) Connectivity & Location	Nodes are loosely connected and geographically dispersed.	Nodes are tightly coupled, usually located in the same data center.	Connectivity through the internet, globally available services.
(vii) Management & Administration	Complex due to multi-organization resource sharing.	Easier as managed by a single organization.	Managed by cloud provider, user handles only configuration.
(viii) Virtualization	Not essential	Not generally used	Heavily relies on virtualization and multi-tenant systems.

2) Explain IaaS model with real world example.

Solution:

IaaS is a cloud computing service model in which users can access and manage fundamental infrastructure resources such as virtual machines, storage, networks, and servers over the internet. It provides a virtualized computing environment on a subscription or pay-as-you-go basis.

In this model:

- The cloud service provider manages the physical infrastructure.
- The customer manages the operating systems, applications, data, and runtime environments.

This allows organizations to avoid the cost and effort of buying, maintaining, and upgrading on-premise hardware.

Real World Example

A common practical example is Amazon Web Services.

A startup wants to launch an online e-commerce

application. Instead of purchasing expensive servers and maintaining a data center, they can use AWS EC2 instances to:

- Host their website,
- Store product data in cloud storage,
- Scale the number of servers during peak shopping periods,
- Protect their website using AWS security services.

They only pay for the computing power and storage they use. When traffic decreases, they scale down to reduce cost.

3) (a) Explain the load balancing in virtualized environment.

(b) Mention how Google Cloud implement load balancing.

Solution:

(a) Load balancing refers to distributing workloads across multiple computing resources so that no single resource becomes a bottleneck. This improves performance, ensures availability, and maximizes resource utilization.

- In a virtualized environment, physical hardware is partitioned into multiple virtual machines via hypervisor.
- Even though virtualization allows efficient use of physical resources, some physical hosts become overloaded while others remain underutilized. Without load balancing, some VMs might become performance bottlenecks, some may idle.
- A load-balancer sits in front of a set of backend VMs. When client requests arrive, the load balancer distributes those requests among backend VMs based on some algorithm (round robin, least connection, etc).

- ⑥ Google cloud is a fully managed load balancing service that distributes traffic across multiple backend resources such as VMs, containers.
- It supports both Layer 4 (TCP/UDP) and Layer 7 (HTTP/HTTPS) load balancing, along with options for external and internal use-cases. Google cloud can be configured at global or regional levels depending on application needs.
 - Key features include autoscaling of backend resources, health checks with automatic failover, and advanced capabilities such as TLS offloading, DDoS protection, CDN integration, and smart routing. When traffic changes, the load balancer dynamically adjusts resource usage and ensures only healthy backends receive traffic.

4) Discuss the layered structure of virtualization in cloud computing.

Solution:

Virtualization in cloud computing is organized into multiple layers, where each layer abstracts and manages specific resources.

(i) Hardware Layer —

- The lowest and foundational layer.
- Consists of physical servers, CPU, RAM, storage devices, network hardware.
- Provides the actual computing resources that are virtualized.
- Cloud providers maintain data centers with large pools of servers and high-speed networks.

(ii) Hypervisor (Virtualization Layer) —

- Responsible for creating, managing, and isolating Virtual Machines by abstracting hardware resources.
- Allocates CPU, memory, and storage to each VM dynamically.
- Type-1 hypervisor runs directly on hardware
- Type-2 hypervisor runs on top of OS

This layer enables multiple VMs to run on a single

physical machine.

(iii) Virtual Machine Layer —

- Contains the actual Virtual Machines that run guest operating systems.
- Each VM behaves like an independent computer, running its own OS and applications.
- Provides full isolation, meaning one VM's failure does not affect others.
- Cloud users deploy their applications here.

(iv) Operating System Layer —

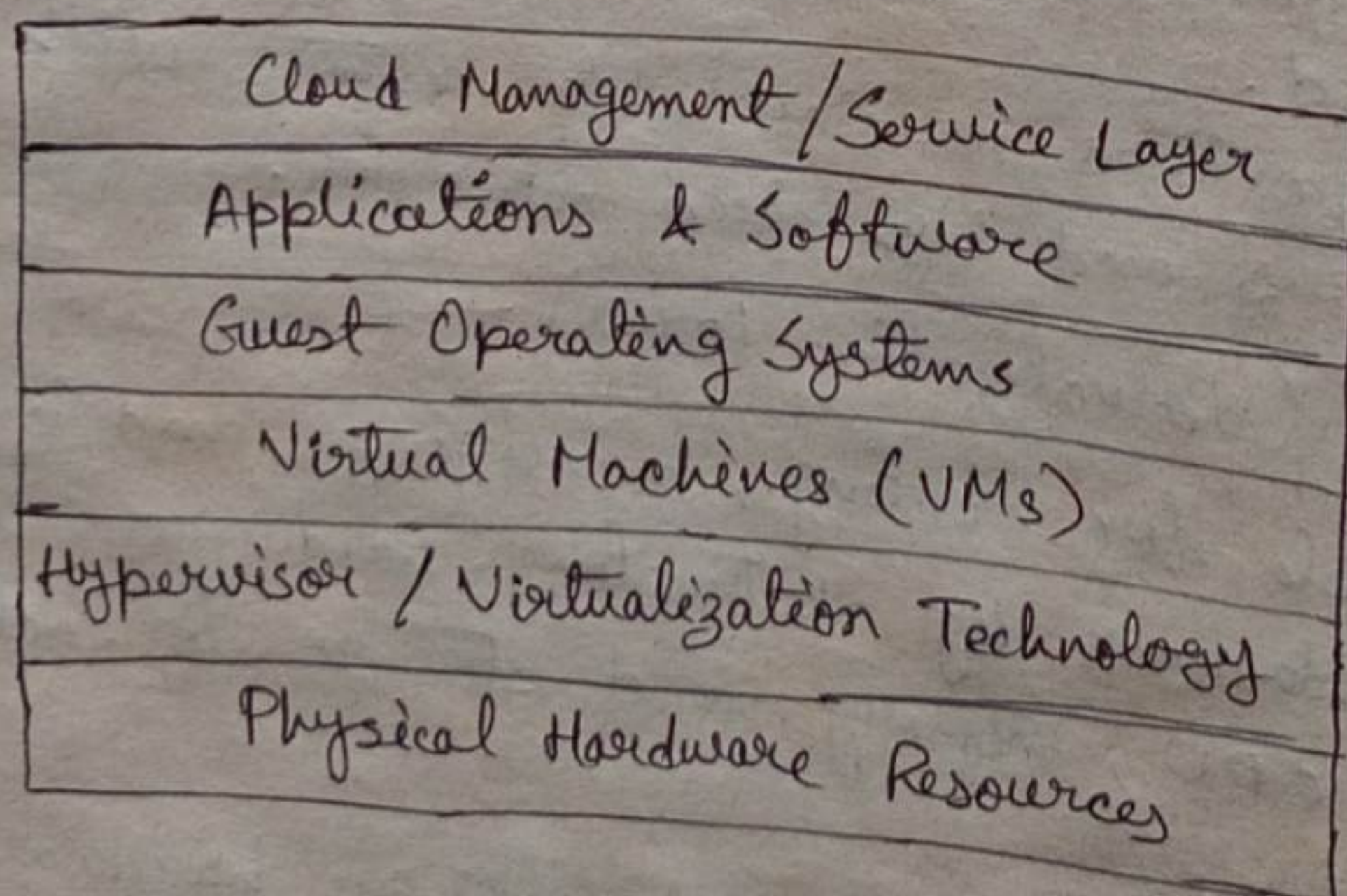
- Every VM has its own guest operating system.
- The OS manages process scheduling, memory, and storage within the VM.
- Some modern cloud services use OS-level virtualization to further optimize resource usage.

(v) Application / Software Layer —

- Contains the end-user applications and services.
- Deployed inside VMs or containers based on the service model.

(vi) Cloud Management & Service Layer —

- Provides automation, billing, security and monitoring tools.
- Ensures the user can interact with the system via dashboards or APIs.



5) Write short note on any two Google Cloud Application.

Solution:

(i) Google Compute Engine (GCE) —

- GCE is an IaaS, it provides virtual machines hosted on Google's infrastructure, letting you run arbitrary operating system and workloads.
- Through Compute Engine you get self-managed VMs, virtual disks, networking capabilities — effectively you get "cloud based servers" instead of on-premises physical servers.
- This is useful when you need full control over the environment.
- It's pay-as-you-go, there's no need for large upfront infrastructure investment. You can scale up or down based on demand, and you only pay for resources you use.

(ii) Google App Engine (GAE) —

- GAE is a fully managed PaaS. Developers can build and deploy web applications or services without managing the underlying servers, OS, or infrastructure. GAE handles those details.
- It supports popular programming languages/runtime environments, and allows developers to focus purely on writing application logic — without worrying about provisioning servers, scaling, load balancing, patching.
- App Engine automatically manages scaling: as traffic increases, GAE scales up instances; if traffic is low, it scales down — giving elasticity. This makes it ideal for web apps, APIs, CRUD based applications, microservices etc.

6) Compare public, private and hybrid community cloud with suitable example.

Solution:

Feature	Public Cloud	Private Cloud	Hybrid Cloud
(i) Ownership	Third party vendor	Single organization	Shared by similar organizations
(ii) Access	Open to public	Restricted	Community members
(iii) Security	Lower than private	Highest	Balanced
(iv) Cost	Lowest pay-as-you-use	High investment	Moderate
(v) Scalability	Very high	Limited to internal resources	Very high
(vi) Control	Low	Full	Partial.
(vii) Example	AWS, Azure	Government Cloud	Hybrid: Azure + On-prem

7) Explain the 6 stages of cloud service life cycle management.

Solution:

Cloud Service Life Cycle Management refers to the complete set of processes involved in planning, developing, deploying, operating and retiring cloud services. The six stages are —

(i) Service Strategy:

- In this stage the organization identifies the business requirements and objectives for using a cloud service.
- Decisions are made about service scope, target users, expected benefits, and cost structure.

(ii) Service Design:

- Defines the architecture and configuration of the cloud service.
- Focuses on capacity planning, networking, storage layout, security, SLA, and compliance.

(iii) Service Development:

- In this stage, the actual implementation and development work is done.
- Coding, virtualization setup, service templates, resource provisioning scripts, and test environments are created.

(iv) Service Deployment:

- The developed service is installed, configured, tested, and released for users.
- Verification of performance, reliability, and user acceptance is carried out.

(v) Service Operation:

- The cloud service is actively used by end-users.
- Includes monitoring, performance optimization, incident handling, support services, and billing management. Ensures continuous availability, reliability and scalability.

(vi) Service Retirement:

- When the service becomes outdated, inefficient, or replaced by a better solution, it is decommissioned.
- Data backup or migration is performed.

8) Describe security issue at the network, host and application level in cloud computing.

Solution:

(i) Network-level security issue — At the network layer, cloud resources are accessed remotely via the Internet, making them vulnerable to attacks during

Transmission.

Key Issues:

(a) Denial of Service (DoS/DDoS) Attacks — Attackers overload network traffic to make cloud services unavailable.

(b) Man-in-the-Middle (MitM) Attacks — Data can be intercepted or altered while moving between users and cloud.

(c) IP Spoofing — Attackers disguise themselves by using forged IP addresses to gain unauthorized access.

(ii) Host-Level Security Issues — Host security refers to protecting the physical servers, hypervisors, and virtual machines on which cloud services run.

Key Issues:

(a) Hypervisor Vulnerabilities — If attackers exploit hypervisor flaws, they can control multiple VMs.

(b) VM Escape Attacks — Malicious code inside a VM can break out and access other VMs or host system.

(c) Resource Sharing Risks — Multi-tenancy can lead to data leakage between tenants.

(iii) Application-Level Security Issues — Applications deployed in the cloud can have vulnerabilities due to coding flaws or insecure API usage.

Key Issues:

(a) Injection Attacks — Attackers insert malicious code to control application or database.

(b) Cross Site Scripting — Scripts are injected into webpages to steal user data.

(c) Insecure APIs — Weak logins or API exploits allow unauthorized access.

Q) Explain the key characteristics of cloud computing highlighting elasticity and scalability.

Solution:

The key characteristics of cloud computing are —

(i) On-Demand Self-Service: Users can provision computing resources automatically without requiring human interaction with the service provider.

(ii) Broad Network Access: Cloud services are accessible over the network from any device using standard web protocols.

(iii) Resource Pooling: Cloud providers use a multi-tenant model where resources are shared dynamically among multiple consumers. The exact physical location of data or server is abstracted from the customer.

(iv) Elasticity: Elasticity refers to the cloud's ability to automatically scale up or down resource capacity based on workload demands.

- Resources expand during peak usage

- Resources shrink during low usage

- Billing is based on actual usage only

Example: E-commerce websites increase server allocation during festival sales and scale back after the sale ends.

(v) Scalability: Scalability is the ability of the cloud system to increase workload capacity by adding more resources without affecting performance.

Two types —

(a) Vertical Scale-Up: Add more power to existing server

(b) Horizontal Scale-Out: Add more server instances

(vi) Measured Service: Cloud systems automatically track usage, enabling pay-as-you-go pricing models.

(vii) Rapid Elasticity and Flexibility: Provisioning of services happens quickly, enabling fast deployment and adaptability to business needs.

10) Differentiate between IaaS, PaaS and SaaS.

Solution:

Feature	IaaS	PaaS	SaaS
(i) Target Users	System admins & IT managers	Developers	End users
(ii) Control Level	Highest	Medium	Lowest
(iii) Management Responsibility	User manages OS, apps, data	User manages apps only	Everything managed by provider
(iv) Access	Virtual hardware	Development tools & environment	Full application
(v) Customization	Highly customizable	Moderate	Minimal
(vi) Example	AWS EC2	Google App Engine	Gmail

11) Discuss scheduling based and transactional based virtualization environment. Provide one example for each.

Solution:

(i) Scheduling Based Virtualization Environment —

(a) In scheduling based virtualization, physical resources such as CPU, memory, and network bandwidth are shared among different VMs based on a scheduling strategy.

(b) The hypervisor schedules CPU time and other resources to VMs, ensuring fair allocation and optimal performance.

(c) Focuses on performance optimization by distributing system resources efficiently among active workloads.

Example:

VMware ESX; or Xen Hypervisor scheduling CPU time among multiple VMs running on the same host.

(ii) Transactional Based Virtualization Environment —

(a) In Transactional-based virtualization, processes

are executed as independent transactions, usually in a containerized or application based environment.

(b) Each transaction is isolated and ensures consistency, integrity, and security, similar to database transaction rules.

(c) Focuses on application level execution rather than hardware or OS level resource sharing.

Example:

• Docker containers running microservices, where each service handles independent transactions.

12) Elaborate the concept of machine imaging in cloud computing.

Solution:

Machine imaging involves creating a complete snapshot of a virtual machine that includes the operating system, installed software, configurations and system settings.

Machine Imaging is important —

- (i) Ensures fast deployment of new systems
- (ii) Allows standardization across all virtual machines in an organization
- (iii) Supports backup and disaster recovery goals
- (iv) Reduces configuration errors and speeds up scaling
- (v) Enables easy migration between cloud environments

Machine Imaging Works —

- (i) A configured virtual machine is created with OS + software + settings.
- (ii) A snapshot of that VM is taken.
- (iii) The image is stored in the cloud as a reusable template.
- (iv) When required, new VMs are launched from that image instantly.

Types of Machine Images —

- (i) Base Image — includes standard OS and minimal configuration
- (ii) Custom Image — includes OS + organization specific configs
- (iii) Golden Image — Fully optimized, secure for replication

Examples —

- Amazon Machine Images in AWS
- Google Cloud Machine Images in Google Compute Engine

Benefits of Machine Imaging —

- (i) Faster provisioning and scaling
- (ii) Reduced deployment cost and effort
- (iii) Ensures consistency and security

Q3) Explain the working Amazon Elastic Blockstore (EBS) and Amazon Relational Database Service (RDS).

Solution:

(i) Amazon Elastic Block Store (EBS) —

Amazon EBS is a block-level storage service used by Amazon EC2 instances.

How it Works —

- (a) When a user launches an EC2 instance, they can attach one or more EBS volumes.
- (b) The volume can store files, databases, OS and application data.
- (c) Even if the EC2 instance stops or terminates, the EBS volume persists and data remain available.
- (d) Snapshots of EBS volumes can be taken for backup and restoration, stored in Amazon S3.

Example —

Running a database on EC2 where data must not be lost when the server is shut down.

(11) Amazon Relational Database Service (RDS) —

Amazon RDS is fully managed relational database service that automates database operations.

How it works —

(a) Users choose a database engine such as MySQL, PostgreSQL, Oracle,

(b) AWS automatically handles:

- Provisioning of servers
- Database configuration & updates
- Storage management

(c) Applications can connect to RDS using standard SQL queries.

Example —

A web application's backend that requires a SQL database with minimal administrative effort.

14) Write short notes on Azure Content Delivery Network, Azure App Fabric.

Solution:

(i) Azure Content Delivery Network (Azure CDN) —

Azure CDN is a distributed network of servers designed to deliver web content ~~such as~~ to users with high speed and low latency.

Key Features:

- Faster content delivery globally through edge servers
- Reduces bandwidth consumption and improves performance
- Enhances website reliability and scalability
- Provides analytics for traffic and performance monitoring

Example:

Streaming video platforms, gaming content delivery, and e-commerce websites use Azure CDN to ensure faster loading for international users.

(ii) Azure App Fabric —

Azure App Fabric is a middleware platform that simplifies the development, deployment, and management of scalable, distributed cloud applications.

Key Features:

- Provides service hosting environment with automatic load balancing
- Supports stateful and stateless microservices
- Ensures high availability through replication and fault tolerance
- Enables easy scaling based on workload

Example:

Used for building high availability enterprise applications such as online transaction systems.

15) Briefly describe the cloud auditing outsourcing life cycle faces.

Solution:

Cloud auditing outsourcing refers to the process where an organization hires a third-party auditor to evaluate cloud services for compliance, security, performance, and governance.

(i) Requirement and Planning Phase —

- Identify objectives of auditing and define scope, audit standards to be evaluated
- Select a trusted third-party auditing organization and establish ~~agg~~ agreements and responsibilities.

(ii) Assessment and Pre-Audit Analysis —

- Gather information about cloud architecture, data flow, and policies
- Understand service provider's SLA and security mechanisms

(iii) Audit Execution Phase —

- Auditor examines access controls, data protection, network security, logging and compliance
- Tools and techniques like penetration testing are performed

(iv) Report Generation Phase —

- Audit findings are documented
- Non-compliance issues, vulnerabilities and risks are highlighted

(v) Remediation and Improvement Phase —

- Cloud service provider and customer take actions to fix reported issues
- Policies and configurations are updated for stronger security and compliance

(vi) Re-evaluation and Continuous Monitoring —

- Cloud services evolve continuously, so periodic re-auditing is required
- Monitoring tools track security status and ensure ongoing compliance.

16) Explain how cloud can be integrated with wireless sensor network. Give two applications.

Solution:

A Wireless Sensor Network (WSN) consists of many small sensor nodes that collect environmental data such as temperature, pressure, motion, and humidity.

How cloud is integrated with WSN:

(i) Data Collection — Sensor nodes gather data from the environment.

(ii) Data Transmission — Data is sent from sensor nodes to a gateway using wireless communication.

(iii) Gateway to Cloud — The gateway connects to the internet and forwards the data to cloud servers.

(iv) Cloud Storage and Processing — Cloud platforms store and analyze the data.

(v) Visualization and Decision Making — Users access real-time dashboards, alerts, and automated responses via web/mobile apps.

Benefits:

- Scalable data storage and computation
- Real time monitoring and analytics
- Reduced cost of on-site infrastructure
- Improved automation and faster decision-making
- Remote access from anywhere

Two applications of Cloud-Integrated WSN:

(a) Smart Agriculture — Sensors collect soil moisture, temperature, and rainfall data.

(b) Healthcare & Patient Monitoring — Wearable sensors record heart rate, oxygen level, and movement.

17) State and explain any 5 benefits of using cloud storage.

Solution:

Cloud storage allows users to store data on remote servers accessed through the internet instead of local physical drives. It offers many operational and economic advantages.

(i) Cost Efficiency: Cloud storage eliminates the need for buying and maintaining expensive hardware. Users pay only for the storage they use, reducing overall IT expenditure.

(ii) Scalability and Flexibility: Cloud storage can be easily expanded or reduced based on requirements.

(iii) Accessibility from Anywhere: Data stored in the

cloud can be accessed from any location at any time using a computer or mobile device with ~~external~~ internet connectivity.

(iv) Data Backup and Disaster Recovery: Cloud service providers maintain multiple copies of data across different data centers.

(v) Enhanced Security: Cloud providers use advanced security techniques such as data encryption, authentication and access control.

10 marks

1) With a suitable diagram explain cloud infrastructure and its architectural framework.

Solution:

Cloud infrastructure refers to the combination of hardware, software, networking resources, and virtualization technologies that work together to deliver cloud computing services over the internet. It provides the foundation for all cloud-based services including IaaS, PaaS and SaaS.

Key components of Cloud Infrastructure

(i) Physical Infrastructure: It provides the basic computing and networking backbone.

(ii) Virtualization Layer: It converts physical resources into virtual resources and it is managed by a hypervisor.

(iii) Cloud Management and Orchestration: It automates resource allocation, monitoring, and service deployment. It also ensures efficient load balancing and resource utilization.

(iv) Service ~~Delivery~~ Delivery Layer: It provides cloud services in three models - IaaS, PaaS and SaaS.

(v) Network Infrastructure: It ensures seamless communication between cloud data centers and end users.

(vi) Security and Compliance Layer: It provides identity management, access control, data encryption and threat monitoring.

Architectural Framework of Cloud Computing

Cloud architecture is generally layered, allowing efficient management and abstraction of resources. The architecture consists of the following five layers:

(i) Physical Layer:

- Contains physical servers, networking, cooling
- Located inside large data centers.

(ii) Virtualization Layer:

- Abstracts physical hardware and creates virtual compute, storage, and network resources.
- Provides isolation between different users.

(iii) Cloud Service Layer:

- Defines service models-

- IaaS → Virtual machines, storage
- PaaS → Runtime environment and development tools
- SaaS → Fully hosted applications for end users

(iv) Application Layer:

- Hosts cloud-enabled software applications accessed by users.

(v) Client Layer:

- Devices and interfaces used by customers to access cloud services.

Benefits

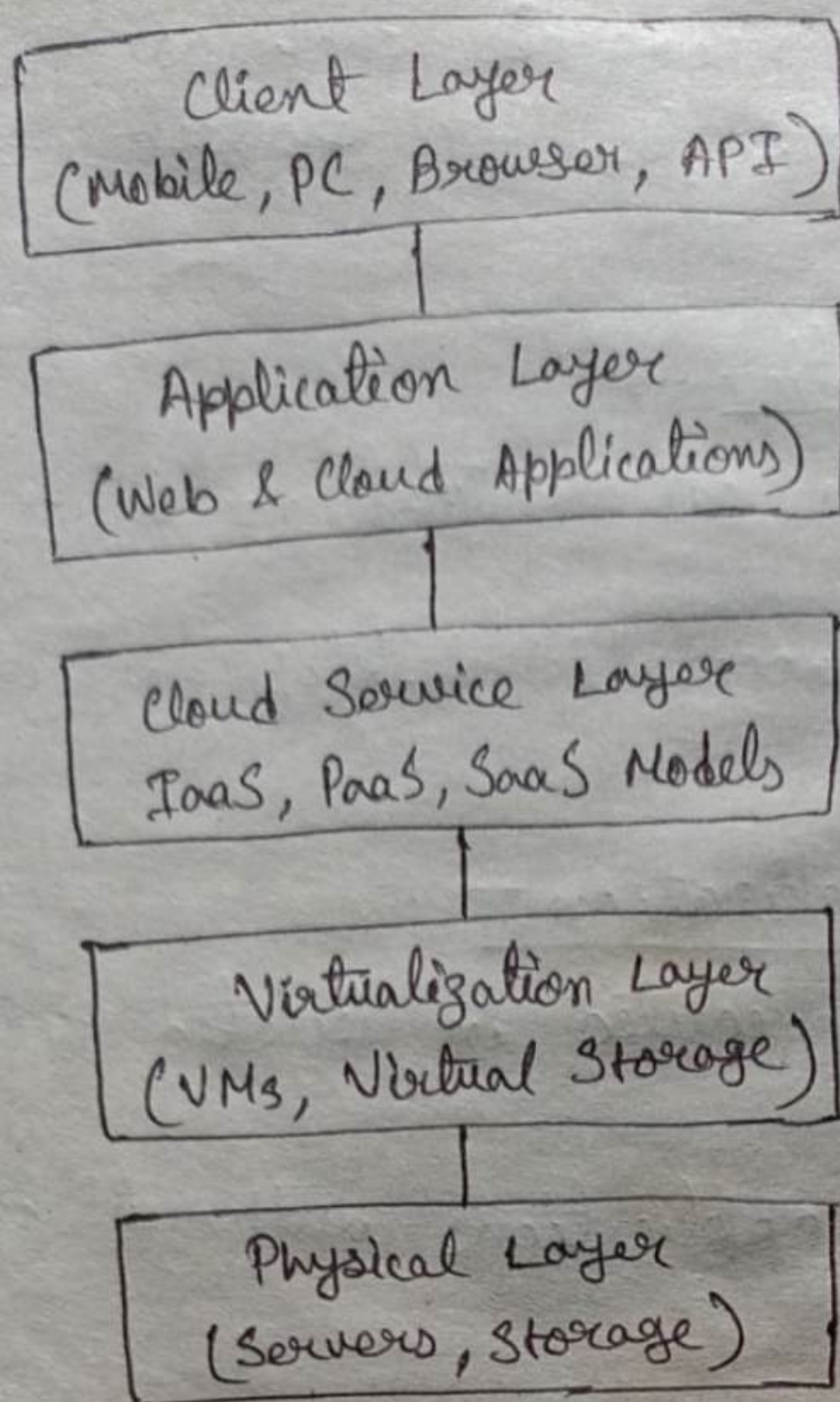
(i) Highly scalable

(ii) Elasticity

(iii) Reduced IT costs

(iv) High Availability & Disaster Recovery

Diagram



2) Discuss different types of virtualization. Compare type 1 and type 2 hypervisor with example.

Solution:

Virtualization is a core technology in cloud computing that allows multiple virtual systems to run on a single physical hardware.

(i) Server Virtualization —

- The physical server is divided into multiple VMs.
- Each VM runs its own OS independently.
- Managed by a hypervisor.
- Used widely in data centers.

(ii) Storage Virtualization —

- Combines multiple physical storage devices into a single logical storage pool.
- Helps in efficient utilization, backup, and disaster recovery.

(iii) Network Virtualization —

- Abstracts network resources such as routers.
- Creates virtual networks for security and optimized communication.

(iv) Desktop Virtualization —

- User desktops run on virtual machines hosted on a server.
- Allows remote access to desktop environments from any device.

(v) Application Virtualization —

- Applications are executed in isolated virtual environments.
- Users can run applications without installing them locally.

(vi) Operating System Virtualization —

- A single OS kernel runs multiple isolated containers.
- Lightweight and ideal for microservices-based cloud applications.

Parameter	Type 1 Hypervisor	Type 2 Hypervisor
(i) Installation	Runs directly on hardware hardware.	Runs on host OS
(ii) Performance	Faster and more efficient	Slower due to extra layer
(iii) Security	High security	Lower security
(iv) Usage	Servers, enterprises	Personal use, small systems
(v) Cost	Mostly commercial	Many free versions available
(vi) Example	VMware ESXi, XenServer	VMware Workstation, Oracle VirtualBox

3) Examine various components of Amazon Web Services such as EC2, S3 and SimpleDB with their practical use.

Solution:

(i) Amazon EC2 —

Amazon EC2 provides resizable virtual compute instances in the cloud. It enables users to launch virtual machines called instances that can run various operating systems depending on application needs.

Key Features:

- Scalability — Increase or decrease the number of virtual machines as demand fluctuates.
- Flexible Configuration — Different instance types optimized for compute, memory, or storage.
- Security — Integrates VPC, IAM roles, key pairs, and security groups.
- Pay-as-you-use pricing.

Practical Uses:

- Hosting web and application servers.
- Running enterprise applications.
- High-performance computing and machine learning workloads.
- Disaster recovery and backup servers.

(ii) Amazon S3 —

Amazon S3 is a highly scalable object storage service that allows users to store and retrieve large amounts of data over the internet.

Key Features:

- Virtually unlimited storage capacity.
- High durability and availability.
- Data stored as objects in buckets.
- Lifecycle policies for automated archival to low-cost

Storage.

- Supports encryption and versioning.

Practical Uses:

- (a) Backup and restore solutions
- (b) Static website hosting
- (c) Big data analytics storage
- (d) Storage for images, videos, logs, and IoT data.
- (e) Content distribution integration with CloudFront.

(iii) Amazon SimpleDB —

Amazon SimpleDB is a NoSQL key-value data store designed for small to medium-scale applications that need semi-structured data with high availability.

Key Features:

- Schema-less database
- Automatic indexing and query support
- Scales automatically with workload
- Designed for applications requiring quick read/write operations.

Practical Uses:

- (a) Lightweight applications needing fast storage and retrieval.
- (b) Shopping cart data storage for e-commerce sites.
- (c) Metadata indexing for media and documents.
- (d) IoT sensor data storage.

4) (a) Explain the 7 step model of cloud migration in detail.

- (b) How do these steps help enterprises in smooth transition in the cloud.

Solution:

a) Cloud migration refers to the process of moving an organization's data, applications, and workloads from on-premises infrastructure to the cloud.

(i) Assessment of Existing System —

It includes:

- Identifying applications, databases, servers, and dependencies.
- Evaluating security, compliance, and network constraints.

(ii) Migration Strategy and Planning —

In this phase, a detailed migration plan is prepared.

- Public, private, or hybrid cloud model.
- Cloud service provider.

(iii) Designing the Cloud Architecture —

This includes:

- Virtual networks, subnets, and routing rules.
- Load balancers and autoscaling groups.

(iv) Application Migration and Development —

In this phase:

- Applications are modified or restructured if required.
- Databases may be ~~redes~~ redesigned to use cloud-native formats.

(v) Data Migration —

It includes:

- Database replication.
- Ensuring data consistency and integrity.

(vi) Testing and Validation —

Tests include:

- Performance testing.
- Integration testing.

(vii) Deployment and Optimization —

This is the final phase, where:

- Applications are deployed onto the cloud.
- Performance monitoring tools are configured.

b) Here's how each step contributes:

- (i) Reduces Risks — Proper assessment and planning help identify compatibility issues, security risks, and dependencies before migration, reducing unexpected failures.
- (ii) Ensures Cost Efficiency — Planning and architectural design prevent overprovisioning and help choose the right cloud services, leading to optimized cost management.
- (iii) Improves Security and Compliance — Security and compliance requirements are properly addressed during assessment, architectural design and testing phases.
- (iv) Minimizes Downtime — Data migration and testing ensure real-time synchronization and prevent major downtime during the transition.
- (v) Enhances Performances — Optimization during the final stage ensures:
 - Autoscaling to handle heavy workloads
 - Load balancing for better availability
- (vi) Supports Business Continuity — Testing for disaster recovery, backup, and failover systems ensures continuous operations even in case of failures.
- (vii) Provides a Clear Roadmap — Each step guides enterprises through a structured, predictable process, making the transition smoother and easier for IT teams and stakeholders.

5) Evaluate the role of Service Oriented Architecture in Modern Cloud Application and Justify how SOA protocol improves workflow coordination in distributed system.

Solution:

SOA plays a major role in shaping how cloud based applications are designed, deployed, and managed.

(i) Promotes Reusability Through Services — Cloud applications

often need repeated functionalities authentication or billing.

(ii) Enhances Scalability and Elasticity — Cloud environments support

dynamic scaling. SOA-based applications are composed of modular services, allowing each service to be scaled independently.

(iii) Ensures Platform and Language Independence —

SOA uses standardized communication ~~pro~~ protocols, which allow services built using: Java, Python, .NET, Node.js

(iv) Supports Distributed Deployment —

Modern cloud applications run across multiple:

- availability zones, regions
- ~~zags~~ virtual machines
- containers
- microservices

(v) Improves Manageability and Monitoring — Each SOA service can

be monitored, logged, and debugged separately to track:

- Service uptime
- API latencies

(vi) Facilitates Integration With External Services —

Modern cloud applications often integrate with third-party services such as:

- Payment gateways
- Authentication providers

SOA protocols help in managing and coordinating complex workflows in distributed cloud systems.

(i) Standardized Communication — SOA protocols provide a standardized mechanism for services running in different environments to communicate reliably.

(ii) Service Discovery and Dynamic Binding — Protocols like UDDI allow services to be registered and discovered dynamically.

(iii) Workflow Orchestration Using BPML — BPML helps coordinate multi-step workflows across different distributed services. It defines:

- (a) Sequence of service calls
- (b) Parallel execution
- (c) Conditional logic
- (d) Error handling

(iv) Loose Coupling Improves Fault Tolerance —

SOA protocols support loose coupling through:

- WSDL-based interface abstraction
- REST's stateless communication

(v) Ensures Security in Distributed Workflows —

SOA protocols support strong security mechanisms like:

- WS-Security
- OAuth / JWT authentication for REST services
- Secure communication via HTTPS

(vi) Supports Transaction Management — In distributed systems, workflows often require multi-step transactions.

SOA protocols support:

- Two-phase commit
- Compensating transactions
- Reliable messaging

6) With a neat diagram explain the cloud reference model, and highlight the shift from traditional IT.

Solutions:

The Cloud Reference Model is a conceptual architecture that describes how cloud services are organized, delivered, and managed. This model consists of five layers arranged from the physical hardware layer to the end-user service delivery layer.

(i) Physical Resource Layer — This is the foundation of the cloud. It includes the actual

hardware installed in data centers.

Components: Physical servers, Network devices etc.

(ii) Virtualization Layer — Virtualization abstracts physical resources and creates VMs,

containers, virtual storage, and virtual networks.

Technologies: VMware ESX; Docker, Kubernetes etc.

(iii) Cloud Service Layer —

① Infrastructure as a Service (IaaS): Provides virtualized

infrastructure like:

- Virtual machines
- Virtual networks

② Platform as a Service (PaaS): Provides platforms for

developers such as:

- Runtime environments
- Middleware

③ Software as a Service (SaaS): Provides end-user

applications accessible via browser or mobile apps.

(iv) Service Orchestration Layer — This layer automates and manages cloud services.

Functions: Resource scheduling, Load balancing etc.

(v) Cloud Service Consumer Layer — This is the top layer where end users

interact with cloud services.

Forms of Access: Web interfaces, Mobile apps, APIs

Users: Developers, Businesses, Individuals

// The difference between traditional IT infrastructure and cloud based systems.

(i) Ownership & Management :

Traditional IT -

- Organizations own and manage their own hardware and software.
- Requires large upfront capital investment.

Cloud -

- Resources are owned and managed by cloud providers.
- Users pay only for what they use

(ii) Resource Provisioning :

Traditional IT -

- Manual provisioning of servers and storage.
- Takes days or weeks to set up infrastructure.

Cloud -

- Automatic provisioning via APIs.
- Resources are available in minutes with elastic scaling.

(iii) Cost Structure :

Traditional IT -

- High upfront cost for hardware and licenses.
- Ongoing maintenance cost.

Cloud -

- Pay-as-you-go model reduces waste.
- No infrastructure maintenance required.

(iv) Scalability and Elasticity :

Traditional IT -

- Limited scaling due to physical constraints.
- Scaling requires purchasing new hardware.

Cloud -

- Near-infinite scalability using autoscaling and

resource pooling.

- Can expand or shrink resources dynamically based on workload.

(v) Application Deployment:

Traditional IT —

- Monolithic applications dominate.
- Deployment cycles are long and complex.

Cloud —

- Supports microservices, DevOps, CI/CD
- Faster updates and continuous delivery

(vi) Reliability and Availability:

Traditional IT —

- Availability depends on local infrastructure.
- Failures lead to downtime

Cloud —

- Multi-region, multi-zone deployment ensures high availability.
- Automatic failover and backup support.

7) Differentiate between virtualization and the traditional approach. Discuss load distribution based and energy aware based environment in virtualization.

Solution:

Difference between Virtualization and Traditional Approach —

Aspect	Traditional Approach	Virtualization Approach
(i) Resource Utilization	Hardware resources are dedicated to a single OS, often underutilized.	Multiple VMs share the same physical hardware, improving utilization.
(ii) Hardware Dependency	Applications depend directly on physical servers and hardware configuration.	Applications run on VMs abstracted from physical hardware through hypervisors.
(iii) Scalability	Scaling requires purchasing and physically installing new hardware.	VMs can be created, cloned, or scaled instantly without new hardware.

Aspect	Traditional Approach	Virtualization Approach
(iv) Cost	High capital expenditure for servers, power, cooling, and maintenance.	Lower cost due to shared resources and pay-as-you-go models in cloud virtualization.
(v) Deployment Time	Slow deployment	Fast deployment
(vi) Fault Tolerance	Hardware failure leads to application downtime.	VM migration, snapshots, and backups ensures minimal downtime.
(vii) Flexibility	Rigid	Highly flexible
(viii) Security	Physical security mainly; fewer layers of isolation.	Strong isolation between VMs; enhanced security configurations.

// Load Distribution Based Environment —

Load distribution ensures that the workload is evenly spread across virtual machines.

Features:

- Dynamic allocation of CPU, memory, ~~band~~ and storage based on workload.
- Ensures balanced performance across VMs.

How it Works:

- (i) Monitoring tools continuously observe resource usage.
- (ii) If a VM is overloaded, tasks are migrated to a less utilized host.
- (iii) Hypervisors use scheduling algorithms.
- (iv) Load balancers help distribute user requests evenly among multiple VMs.

Advantages:

- (i) High performance and responsiveness.
- (ii) Better utilization of resources.
- (iii) Reduce chances of system failure due to overload.

Energy Aware Based Environment

Energy aware virtualization focuses on minimizing power consumption while maintaining performance.

How it works:

- (i) Idle or low-use VMs are migrated to a few physical servers.
- (ii) CPU frequency is reduced during low workloads.
- (iii) Workloads are assigned to machines that operate efficiently.
- (iv) Servers automatically shift to sleep mode when not needed.
- (v) Temperature-aware VM placement reduces cooling requirements.

Benefits:

- (i) Lower operational cost due to reduced electricity usage.
- (ii) Increased lifespan of hardware as fewer servers run at high load.
- (iii) Reduced carbon footprint, supporting green computing.

8) Analyze how SOA and its protocol stacks enable workflow and coordination across cloud services.

Solution:

Service Oriented Architecture (SOA) is a design framework where software components are provided as loosely coupled, reusable services that communicate through standardized protocols.

SOA helps cloud systems by breaking down applications into independent services that can be:

- (i) Executed ~~on~~ remotely
- (ii) Scaled independently
- (iii) Reused by multiple applications

SOA uses a layered protocol stack to enable communication, description, coordination, and security. The key protocol layers are —

① Transport Protocol Layer: Handles low-level communication

between cloud services.

Common protocols: HTTP/HTTPS, SMTP

Role:

- Ensures services communicate reliably over the internet
- Provides standardized path for request-response messages

(b) Message Protocol Layer: Defines message formats for communication between services.

Common Protocols: SOAP, REST, JSON, XML

Role:

- Structures the data exchanged between services
- Enables asynchronous and synchronous workflows

(c) Service Description Layer: Defines how a service is described and advertised.

Key Technologies: WSDL, OpenAPI

Role:

- Provides machine-readable descriptions
- Enables dynamic discovery of services

(d) Service Discovery & Registry Layer: Handles registration and lookup of services.

Role:

- Improves scalability by balancing which service instance to call

(e) Service Workflow & Coordination Layer: This is the layer that binds services into complete business processes.

Role: - Supports execution of long-running processes

(f) Quality of Service (QoS) and Policy Layer: Ensures security, reliability, and performance across distributed cloud services.

Role:

- Maintains secure communication

SOA Protocol Stacks Enable Workflow Coordination in cloud Environments —

(i) SOA protocols like REST, XML and JSON are universally supported, enabling seamless communication between AWS, Azure, and GCP.

- (ii) SOA registries allow services to be dynamically located, enabling cloud autoscaling and automatic failover.
- (iii) Message protocols carry workflow data among services, allowing distributed tasks to be coordinated.
- (iv) Protocols ensure atomic, consistent operations for banking, payments, or cloud billing systems.
- (v) Security protocols protect workflows that span multiple cloud regions or providers.
- (vi) Workflow engines use service description and coordination protocols to connect multiple cloud services.

9) Discuss the ~~need~~^{need} for ~~live~~^{live} migration of virtual machine and the design process. Highlight possible security issue during migration.

Solution:

Need for Live Migration of VMs

Live migration is essential for maintaining performance, availability, and resource efficiency in cloud data centers.

(a) Load Balancing — When a physical host becomes overloaded with CPU, memory, or I/O demands, live migration helps shift VMs to under-utilized servers, ensuring optimal performance.

(b) Hardware Maintenance — Live migration enables proactive maintenance of a server by moving VMs without downtime.

(c) Energy Efficiency — During low workloads, VMs can be moved to fewer active servers, allowing idle machines to be powered down.

(d) Disaster Avoidance — If system operators detect risks, VMs can be migrated quickly to safer nodes.

(e) Resource Optimization — Ensures better utilization of cloud resources by distributing VMs based on performance, memory availability.

Design Process of VM Live Migration

Live migration typically follows a structured process to minimize downtime and data loss.

- (i) Pre-Migration: Source and destination hosts are selected based on current load and resource availability.
- (ii) Reservation of Resources: The destination host reserves CPU, memory, storage access, and network resources required for the incoming VM.
- (iii) Memory Pre-Copy Phase: VM continues to run on the source host.
- (iv) Stop and Copy: The VM is paused for a very short time.
- (v) VM Resume on Destination: The VM is resumed on the destination server using the transferred state.
- (vi) Post-Migration Clean-Up: Resources on the source host are released.

Possible Security Issues During VM Migration

- (a) Data Interception: Migration data contains sensitive contents. If it is sent over an insecure channel, attackers may intercept and read the data.
- (b) Man-in-the-Middle Attacks: An attacker can impersonate the destination host and capture the VM's memory and CPU state.
- (c) VM Hijacking: If authentication and authorization are weak, attackers may trigger unauthorized migration or ~~also~~ redirect a VM to a malicious server.
- (d) Replay Attacks: An attacker may capture migration data and replay it to corrupt VM state or gain access.
- (e) Denial of Service (DoS) Attacks: Attackers may overwhelm the migration network path.

④ Resource Exhaustion Attacks: Malicious processes may continuously alter memory pages, making migration impossible.

10) Analyse the challenges of ensuring data security, privacy and compliance in public cloud selling. Provide recent realistic example to support the analysis.

Solution:

Using a public cloud offers ~~may~~ many benefits, but it also introduces significant risks when it comes to protecting data.

(i) Shared Responsibility & Loss of Direct Control:

In public cloud, infrastructure is managed by the cloud provider - not the customer. This means customers may have limited visibility into how data is stored, processed, secured or backed up inside the provider's environment.

(ii) Data Privacy:

Public cloud providers often store data across data centres in multiple geographic regions. This raises legal and regulatory compliance challenges.

(iii) Misconfiguration, Weak Authentication & Access Control:

Many data breaches in public clouds stem from misconfigurations or weak/absent authentication methods.

(iv) Complex Compliance Requirements & Auditability:

Enterprises - especially in regulated sectors - must comply with strict regulatory frameworks.

(v) Insider Threats and Multi-tenancy Risks:

Public cloud environments are multi-tenant: many customers share underlying hardware and services. This raises the risk that a malicious insider - or a compromised tenant - could exploit vulnerabilities to access or leak data belonging to other tenants.

Recent Realistic Example

- A strong recent example is the 2024-2025 breach affecting customers of Snowflake - a major cloud-based data warehousing platform. Reports indicate that a large-scale cyber-attack compromised sensitive customer data across many tenant accounts in Snowflake environments.
- This breach underscores how even leading cloud vendors and their customers can be ~~not~~ vulnerable - particularly when there are misconfigurations, ~~or~~ insufficient access control, or insufficient monitoring.
- Another recent trend: In 2025, a security researcher uncovered more than 17000 secrets exposed in public repositories on a popular cloud-based code hosting service. Many of these were linked to public cloud accounts, illustrating how human error + ~~mis~~ mismanagement can lead to major cloud security risks.
- Such incidents reveal that public cloud does not automatically guarantee ~~a~~ security or compliance - it depends heavily on how customers configure, manage and ~~and~~ monitor their cloud environment.

11) Evaluate the effectiveness of public vs. private cloud storage for a large university environment. And justify which approach is better in terms of scalability, security and cost.

Solution:

The comparison between public and private cloud storage-

A large university typically stores:

- Student records, grades
- Research data
- Course materials, video lectures

Dimension	Public Cloud Storage	Private Cloud Storage
(i) Ownership	Provider-owned infrastructure.	University-owned or hosted in dedicated on-premises datacenter.
(ii) Scalability	Virtually unlimited, elastic on demand	Limited by physical capacity unless large CAPEX invested.
(iii) Cost Model	It is pay-as-you-go model.	Cost includes buying hardware, power, cooling, staff; predictable fixed costs
(iv) Security / Control	Strong provider controls but less direct physical control; multi-tenant	Full control over hardware, network, policies; single-tenant
(v) Compliance & Data Sovereignty	Providers offer compliance features / regional controls	Easier to demonstrate physical control & data locality
(vi) Time-to-provision	Minutes to hours	Days to weeks
(vii) Management Overhead	Lower for infra; provider manages hardware	Higher

(i) Scalability :

Public cloud is ~~clearly~~ ~~at~~ superior for on-demand scalability and elasticity. If the university expects large variability, public cloud makes scaling trivial.

(ii) Security & Privacy :

Private cloud offers stronger control by default. Public cloud can be sufficiently secure if the university has strong governance, correct configuration, encryption, KMS policies, and monitoring. For many universities, a mixed approach (public + private) balances risk and capability.

(iii) Cost:

No universal winner. Small-to-medium, variable workloads → public cloud favored. Very large, predictable steady-state storage → private cloud can be cost-effective if the university can achieve high utilization.

So, recommended to use hybrid strategy.

- (a) Balances scalability with control / security.
- (b) Optimizes costs by keeping predictable sensitive workloads on private infrastructure while leveraging public cloud economies for spikes and archival.
- (c) Provides flexibility for research groups with different needs.

12) Elaborate on a non relational data model in the cloud with mongodb as an example. How does the cap theorem apply here.

Solution:

Non-relational data models are designed to overcome the limitations of relational databases, especially in distributed and large-scale cloud environments. They support:

- Schema flexibility
- Horizontal scalability
- High availability and distributed storage

MongoDB as an Example of a Cloud-Based NoSQL Database—

MongoDB is a document-oriented NoSQL database that stores data in BSON (Binary JSON) format, making it flexible and capable of storing complex structures.

(a) Document-Oriented Storage:

Data is stored in JSON-like documents. This makes MongoDB perfect for cloud applications with dynamic data structures.

(b) Schema Flexibility:

Different documents within the same collection can

have different fields. This is ideal for cloud-native applications with evolving data requirements.

③ Horizontal Scalability:

MongoDB distributes data across multiple nodes using a sharding key, enabling massive horizontal scaling - crucial for cloud systems that handle large user bases.

④ Replication via Replica Sets:

MongoDB uses replica sets to maintain multiple copies of data across various nodes.

⑤ Cloud-Native Deployment:

MongoDB Atlas lets organizations deploy MongoDB clusters across cloud platforms.

|| The CAP theorem states that in a distributed system, it is impossible to simultaneously provide all three guarantees:

- C - Consistency
- A - Availability
- P - Partition Tolerance

MongoDB can operate as both CP (Consistency + Partition Tolerance) or AP (Availability + Partition Tolerance) depending on how it is configured.

(i) CP Mode

MongoDB ensures consistency by:

- Using a single primary node for ~~all~~ writes
- Rejecting writes if the primary is lost

Behaviour:

If the primary node goes down, MongoDB pauses writes until a new primary is elected. This ensures strong consistency, but reduces availability temporarily.

(ii) AP Mode

MongoDB can be tuned for higher availability using:

- Read preference
- Write concern

This allows operations even if some nodes are unreachable.

Behavior:

Reads may return slightly outdated data, but the system remains available.